System Requirements Specification (SRS)

Project: Labaid InsureTech Platform

Version: 2.2

Date: December 2025

Status: Final Draft

Control Level: A

Revision History

SL	Date	Revised By	Description
01	12/06/2025	Faruk Hannan	Initial V2.0 with novel ideas
02	15/12/2025	AI Assistant	V2.1 Complete - Enhanced with professional formatting
03	16/12/2025	Faruk Hannan	V2.2 Complete - MD feedback integrated, comprehensive tables added

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1. Introduction

1.1 Purpose

This System Requirements Specification (SRS) documents comprehensive system-level requirements for the LabAid InsureTech platform, comprising mobile applications, partner portals, admin portal, and backend services. This specification serves as the authoritative technical documentation for design, development, integration, testing, deployment, and operations teams, ensuring compliance with Bangladesh Insurance Development and Regulatory Authority (IDRA) regulations and local regulatory frameworks.

1.2 Scope

The system will enable digital onboarding, product discovery, policy purchase, digital KYC, payment processing, claims submission and tracking, partner integrations, and admin/insurer workflows. This SRS covers Phase 1 (core digital capabilities) while flagging Phase 2/3 enhancements (AI underwriting, Voice-aided guidance, IoT based connectivity) for future releases. See Business Plan for strategic context.

1.3 Local Challenges

With a low penetration rate of insurance compared to other countries in the region and considering mass literacy level and lack of awareness about insurance, a very simplified onboarding flow with step-by-step explanations and visual cues are required for users in the UI/UX space. Also considering rural internet network constraints, overall speed (minimum 3G), minimum specifications of edge devices, and lack of availability of local cloud storage support within the country for reducing latency, specific optimization strategies are essential for Bangladesh market penetration.

1.4 Definitions, Acronyms & Abbreviations

Term	Definition
IDRA	Insurance Development & Regulatory Authority (Bangladesh)
KYC	Know Your Customer
KYB	Know Your Business
FCR	Financial Condition Report (IDRA requirement)
CARAMELS	Capital Adequacy, Reinsurance Arrangements, Management, Earnings, Liquidity & Asset quality, Sensitivity
SAR	Suspicious Activity Report (AML requirement)
STR	Suspicious Transaction Report
BFIU	Bangladesh Financial Intelligence Unit
MFS	Mobile Financial Service (bKash, Nagad, Rocket)
API	Application Programming Interface
EHR	Electronic Health Record
ZHCT	Zero Human Touch Claims
UBI	Usage Based Insurance
IAM	Identity and Access Management
ACL	Access Control List
RBAC	Role Based Access Control
ABAC	Attribute Based Access Control
UAT	User Acceptance Testing
MOU	Memorandum of Understanding
TAT	Turn Around Time
RACI	Responsible, Accountable, Consulted, Informed
RFC	Request For Change

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2. Market Context

2.1 Bangladesh Insurance Market Challenges

- **Low insurance penetration** compared to regional countries
- **Mass literacy level** and lack of awareness about insurance products
- **Rural internet network constraints** (Minimum 3G, EDGE not supported)
- **Minimum specification edge devices** (4GB RAM minimum support)
- **Lack of local cloud storage** support requiring optimization strategies
- **Frequent network switching** requiring resilient architecture

2.2 Target User Categories

Customers: Mobile-first, diverse digital literacy

- **Type 1: Urban and Digitally Literate** → Self-service mobile app users
- **Type 2: Semi-Urban/Agent-assisted** → Payment with agent support
- **Type 3: Rural/Agricultural** → Voice-assisted workflow required

Stakeholders: **Broader classification for system access control**

- **Partners:** System-onboarded, collaborative organizations (MFS, hospitals, e-commerce)
- **Insurer Agents:** Staff at partner organizations under ACL
- **Insurer Underwriters:** Internal to partner's organization insurers (API consumers)
- **Independent Agent:** Onboarded, certified payment agents in remote locations (dashboard)
- **Vendor:** Independent service provider to InsureTech as a client (dashboard)

Administrative Roles Hierarchy:

- **System Admin:** Root level to cloud provider gateway (IAM, ACL, Policy)
- **Repository Admin:** Root level to repo and droplet (Pull, Merge, Deploy access)
- **Database Admin:** Root level to databases/storage (Full management access)
- **Business Admin:** Root level to portal (policy management, product updates)
- **Focal Person:** Root level to Partner management (partner verification, onboarding, dispute resolution) - **KEY BRIDGE ROLE**
- **Partner Admin:** Partner-side root tenant (IAM, ACL control)
- **Dev:** User level (software development)
- **Support/Call Centre:** Assisted onboarding & claims help (dashboard)

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3. Overall Description

3.1 Product Perspective

The platform is **cloud-native, secure-first, microservices-based, event-driven, and mobile-friendly**.

System Architecture Components:

- **Mobile apps (iOS/Android)** — Customer experience with Bangladesh-specific UI/UX considerations
- **Partner/Agent Portal** — Web-based onboarding & embedded flows
- **Admin Portal** — Product, pricing, user and claims management with regulatory reporting
- **Backend Services** — Authentication, authorization, partner/agent management, policy engine, profiling service, contract management, payment gateway adapters, risk management, fraud detection, renewal service, transactional debit-credit service, storage service, LLM multi-agent network, AI assistant service, MCP servers, IoT-broker service, orchestration service (Kafka), notification service, ticketing service (customer support), analytics, reporting
- **Third-party Integrations** — Insurer APIs, LabAid systems, MFS/telco APIs, regulatory portals

3.2 User Classes & Characteristics

User Class	Characteristics	Digital Literacy	Primary Access Method
Primary Customers	Urban professionals, middle-class families	High	Mobile App (Primary)
Secondary Customers	Rural farmers, small business owners	Low to Medium	Mobile App with voice assistance
Agent/Partners	MFS agents, hospital staff, e-commerce representatives	Medium to High	Partner Portal
Insurer Underwriters	Internal insurance company staff	High	API Integration
Admin Users	Business administrators, product managers	High	Admin Portal
Support Staff	Call center operators, customer service	Medium to High	Admin Portal (limited access)
Compliance Officers	Regulatory compliance staff	High	Admin Portal (compliance module)
Focal Person	Partner bridge, verification, dispute resolution	High	Admin Portal (partner management)

3.3 Operating Environment

Cloud Infrastructure:

- **Primary:** AWS/Azure Bangladesh region or IDRA-compliant data centers
- **Backup:** Multi-region disaster recovery setup
- **Compliance:** Data residency requirements per Bangladesh regulations
- **Latency:** Maximum 80ms added latency for regional deployments

Mobile Platforms:

- **iOS:** Version 13.0 and above
- **Android:** Version 9.0 (API level 28) and above
- **Minimum support:** 4GB RAM devices
- **Network:** Minimum 3G (EDGE will not be supported) with frequent network switching

Mobile App Constraints:

- **App Download:** Lowest download size as low as 10 Megabyte, gradual download and caching of data run up to maximum 100 Megabyte
- **Offline Capabilities:** Minimum persistent view of critical data
- **Low Bandwidth Mode:** Minimum feature with device level cached static data
- **Low Resource Mode:** Must run with minimum memory and low power mode

Web Browsers:

- Chrome, Brave, Firefox, Edge (latest two versions)

Fallback Mechanisms:

- **USSD/SMS fallbacks:** Retry and acknowledgement support

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4. System Features & Functional Requirements

Each requirement has a **Function Group unique ID (FG-XXX)** and **Functional Requirements unique ID (FR-XXX)**.

Functionality Deploy Phases: Phase 1, Phase 1.5, Phase 2, Phase 3

Priority levels:

- **M** = Mandatory (Phase 1)
- **D** = Desirable (Phase 1.5)
- **F** = Future (Phase 2/3)

4.1 Authentication & User Management

FG	ID	Requirement Description	Priority
FG-001		Authentication & User Management	M
	FR-001	The system shall allow Customers to register and login via mobile number with OTP verification with minimum data (name, date of birth, mobile no, email optional)	M
	FR-002	The system shall auto update OTP from SMS with one time read permission from device	D
	FR-003	The system shall have SMS OTP failure fallback with persistent form state, retry loop and resend button	M
	FR-004	The system shall have image capture step for user verification step 1	M
	FR-005	The system shall perform login step via e-KYC as per Office of the Controller of Certifying Authorities Bangladesh	D
	FR-006	The system shall automate with voice-assisted workflow on landing page for Type 3 customer to guide to login	F
	FR-007	The system shall have option for logged in customer to link with social account via OAuth2 (for Type 1, Type 2 customer: Google, Facebook) to finalize profile	D
	FR-008	The system shall have mandatory NID upload for user verification step 2	M
	FR-009	The system shall digitize and process NID data	D
	FR-010	The system shall show green check mark for user verification step	D
	FR-011	The system shall have option of InsureTech employed Focal Person to onboard a pre-approved, MOU-signed partner on behalf of the partner, setup basic ACL, and provide one Partner Admin with temporary password	M
	FR-012	The system shall have stakeholders registration via OPEN ID Identity provider	M

	FR-013	The system shall have stakeholders registration via SAML Identity provider	D
	FR-014	The system shall have stakeholders registration via SSO provider	D
	FR-015	The system shall have stakeholders registration via email with email verification	M
	FR-016	The system shall lock verified stakeholder KYB data from unauthorized updates	M

4.2 Authorization & Access Control

FG	ID	Requirement Description	Priority
FG-002		**Authorization & Access Control**	M
	FR-017	The system shall implement Role-Based Access Control (RBAC) for all stakeholder types with granular permission sets	M
	FR-018	The system shall implement Attribute-Based Access Control (ABAC) for context-aware authorization decisions based on user attributes, resource attributes, and environmental conditions	D
	FR-019	The system shall maintain Access Control Lists (ACL) for fine-grained resource-level permissions	M
	FR-020	The system shall enforce hierarchical access control where System Admin > Repository Admin > Database Admin > Business Admin > Focal Person > Partner Admin > Dev > Support	M
	FR-021	The system shall allow Focal Person exclusive rights to partner verification, onboarding, and dispute resolution workflows	M
	FR-022	The system shall enable Partner Admin to manage their tenant's IAM and ACL independently within their scope	M
	FR-023	The system shall implement policy-based authorization for microservices using policy engine (OPA or equivalent)	D
	FR-024	The system shall maintain separation of duties (SoD) rules preventing conflicting role assignments	M
	FR-025	The system shall support temporary privilege escalation with time-bound access and audit logging	D
	FR-026	The system shall implement mandatory 2FA for all admin-level access (System Admin, Business Admin, Database Admin, Repository Admin, Focal Person)	M

4.3 Product Catalog & Policy Discovery

FG	ID	Requirement Description	Priority
FG-003		**Product Catalog & Policy Discovery**	M
	FR-027	The system shall maintain product catalog API: Serve product list with metadata (name, coverage, premiums, insurer, T&Cs)	M
	FR-028	The system shall provide filter & compare: Support filtering (category, premium, coverage) and side-by-side comparison (up to 3). UI screens show these flows	M
	FR-029	The system shall display product detail: Full policy wording, exclusions, and illustrative premium breakdown	M
	FR-030	The system shall load catalog in less than 2 seconds for optimal user experience	M
	FR-031	The system shall support product search in both Bengali and English languages	M
	FR-032	The system shall implement intelligent caching for product catalog with 5-minute TTL	M

4.4 Policy Purchase & Issuance

FG	ID	Requirement Description	Priority
FG-04		**Policy Purchase & Issuance**	M
	FR-033	The system shall implement multi-step purchase flow: Personal details, nominee, document upload, review, payment, confirmation. Flow follows provided mockups	M
	FR-034	The system shall provide premium calculation: Request insurer API or local pricing engine and show breakdown	M
	FR-035	The system shall support payment integration: Support bKash, Nagad, Rocket, card (PCI-DSS compliant), bank transfer	M
	FR-036	The system shall implement policy issuance: On successful payment, generate a digital policy certificate (PDF) and store in user account	M
	FR-037	The system shall support promo/discounts: Support coupon codes and partner-discount flows	D
	FR-038	The system shall support promo code for product discount	M

	FR-039	The system shall maintain purchase workflow state for recovery in case of network interruption	M
	FR-040	The system shall send purchase confirmation via SMS, push notification, and email	M

4.5 Claims Management

FG	ID	Requirement Description	Priority
FG-05		Claims Management	M
	FR-041	The system shall have a form of fixed steps of Insurance Claims - Policy prefill, claim reason selection, document upload (images, bills)	M
	FR-042	The system shall show success screen after initial screening of documents after submission	M
	FR-043	The system shall make a digital hash to identify each unique submission and keep as internal record	M
	FR-044	The system shall automatically notify partner for new claims and start common status dashboard with customer	M
	FR-045	The system shall show status tracking of the claim in the dashboard to customer	M
	FR-046	The system shall have a chat interface with customer and partner agent in dashboard	D
	FR-047	The system shall have a WebRTC based video call support with customer and partner agent in dashboard	D
	FR-048	The system shall have agents note support in dashboard	M
	FR-049	The system shall allow partner to add verification details for InsureTech record keeping and agree on Approval Matrix timeline	M

	FR-050	The system shall allow partner to make initial approval or rejection request to InsureTech	M
	FR-051	The system shall allow Business Admin and Focal Person joint approval or rejection upon internal verification	M
	FR-052	The system shall automate payment process as per customer selected payment channel	M
	FR-053	The system shall have auto verification, claim approval and payment for small claims with pre-agreement as per Partner, Insurer and InsureTech	D
	FR-054	The system shall have auto fraud detection system to detect frequent claims and changed uploads by customer	D
	FR-055	The system shall provide system warning to customer as per InsureTech policy	M
	FR-056	The system shall auto revoke access to customer as per InsureTech policy for suspicious activity	D
	FR-057	The system shall maintain proper balance sheet on Customer, Partner, Agent and InsureTech level for various selected period of time	D
	FR-058	The system shall implement tiered approval workflow based on claim amount as per defined Approval Matrix	M

Claims Approval Matrix:

Claimed Amount	Approval Level	Approver(s)	Maximum TAT
BDT 0-10K	L1 Auto/Officer	System Auto-Approval OR Claims Officer	24 Hours
BDT 10K-50K	L2 Manager	Claims Manager	3 days
BDT 50K-2L	L3 Head	Business Admin + Focal Person (Joint)	7 days
BDT 2L+	Board	Board + Insurer Approval	15 days

4.6 Policy Management & Renewals

FG	ID	Requirement Description	Priority
FG-006		Policy Management & Renewals**	M
	FR-059	The system shall provide customer a policy dashboard after first product purchase	M
	FR-060	The system shall show active and past policies, renewal prompts, policy PDF download in customer dashboard	M
	FR-061	The system shall provide customer two options for unpaid policies renewal: auto and manual	M
	FR-062	The system shall guide customer in clear steps for manual policy renewal	M
	FR-063	The system shall allow customer to partially adjust policy data: 1. Current address 2. Nominee	M

4.7 Notifications & Communication

FG	ID	Requirement Description	Priority
FG-007		Notifications & Communication	M
	FR-064	The system shall have Kafka (or equivalent) event-driven notification and orchestration system triggering following channels like: In-System PUSH (portal & mobile), SMS, Email	M
	FR-065	The system shall auto trigger notification for verification, purchase confirmation, claims updates, renewal reminders	M
	FR-066	The system shall allow user to be notified for additional events as per provided list (e.g., new product launch)	D
	FR-067	The system shall provide partners to create secondary marketing notification for customers by filtering age, sex, groups	D
	FR-068	The system shall allow customers with opt-in/opt-out policy for additional notification	M
	FR-069	The system shall support customer mobile mute mode with minimum text notification (avoiding on-system push for edge devices)	M
	FR-070	The system shall allow customer support call within app	D
	FR-071	The system shall provide common Q&A within app for customer knowledge	M
	FR-072	The system shall provide AI chatbot for customer support and automatic ticketing for unresolved issue	F
	FR-073	The system shall provide common question and answer form for customer support call center (Vendor)	D
	FR-074	The system shall provide ticket form fill up option with auto recorded customer support call	D

4.8 Admin & Reporting

FG	ID	Requirement Description	Priority
FG-008		Admin & Reporting	M
	FR-075	The system shall maintain separate dashboard for different admin as per defined role, resources, policies by InsureTech management	M
	FR-076	The system shall maintain strict 2FA for any admin level access	M
	FR-077	The system shall make dynamic content view based on context, workflow, and data access available for different admin	M
	FR-078	The system shall provide common dashboard for domain level, team level	D
	FR-079	The system shall allow internal approval, task issuing to internal users	D
	FR-080	The system shall have separate user management tab	M
	FR-081	The system shall have separate product management tab	M
	FR-082	The system shall have separate claim management tab	M
	FR-083	The system shall have assigned task tab per user	M
	FR-084	The system shall have option of reporting as per access to users	M
	FR-085	The system shall have minimum reporting options as: Daily sales, claims ratio, partner performance, policy counts and KPIs (aligned to business plan targets)	M

4.9 Partner Portal

FG	ID	Requirement Description	Priority
FG-009		Partner Portal (Hospitals, E-commerce)	M
	FR-086	The system shall provide special dashboard for hospital partner to be able to initiate purchase insurance for any customer on behalf of them	M
	FR-087	The system shall have option to transfer customer record with partner provided API and purchase ID and authentication token	D
	FR-088	The system shall provide API for E-commerce partner to embed product in page and special dashboard to purchase product from InsureTech	M
	FR-089	The system shall provide sandbox for 3rd party developer for authentication and testing payment and purchase process	F

	FR-090	The system shall provide any partner onboarding analytics, leads, commission statements in their dedicated dashboard	M
	FR-091	The system shall have internal Admin special data product as Business Intelligence Tool: TBD (Options: Metabase, Tableau, Power BI, custom dashboards). Data Source: Read replica of production database. Update Frequency: Near real-time (5-minute lag acceptable). Dashboards: Executive (Daily sales, policy count, claims ratio, revenue), Operations (System health, support tickets, processing times), Compliance (AML flags, IDRA report status, audit logs)	F
	FR-092	The system shall provide any partner onboarding analytics, leads, commission statements via API	M

4.10 AI & Automation Features

FG	ID	Requirement Description	Priority
FG-010		**AI & Automation Features**	F
	FR-093	The system shall have purpose built AI Chatbot to assist customer during product search, selection, purchase, verification, payment stage	F
	FR-094	The system shall have LLM multi-agent network for intelligent document processing and validation	F
	FR-095	The system shall implement AI-powered fraud detection using pattern recognition	F
	FR-096	The system shall support predictive analytics for risk assessment	F

4.11 Audit & Logging

FG	ID	Requirement Description	Priority
FG-011		Audit & Logging	M
	FR-097	The system shall maintain immutable logs for critical actions such as: policy issue, claim approval, claim rejection, payment and dispute	M
	FR-098	The system shall maintain data retention policy up to minimum of 20 years to maintain records for regulatory compliance	D
	FR-099	The system shall track each logged in user for auxiliary actions and will have additional data logs	D
	FR-100	The system shall allow partner to maintain additional data logs as per customer and InsureTech MOU	F
	FR-101	The system shall provide special portal to regulatory body to access requested data as per policy and law of regulatory bodies	M

4.12 User Interface Requirements

FG	ID	Requirement Description	Priority
FG-012		User Interface*	M
	FR-102	The system shall maintain similar user interface with different operating systems	M
	FR-103	The system shall provide smart data widget for mobile user	D
	FR-104	The system shall provide voice assisted workflow for Type 3 user	F
	FR-105	The system shall provide desktop first web UI for portal	M
	FR-106	The system shall take minimum permissions for all services from user device (e.g., camera, one time message read)	M

4.13 API Design and Data Flow

FG	ID	Requirement Description	Priority
FG-013		API Design and Data Flow	M
	FR-107	The system shall maintain two levels of API: private and public	M
	FR-108	The system shall have three levels of private API as Category 1, Category 2, and Category 3 with secure middle layer	D
	FR-109	The system shall use Protocol Buffer and gRPC based communication for Category 1 API between gateway and microservices in HTTP with system admin middle layer	F
	FR-110	The system shall use GraphQL based API as Category 2 for gateway and customer device with JWT token system and OAuth v2 based security middle layer	M
	FR-111	The system shall use RESTful API with JSON with compliance of OPEN API standard as Category 3 for 3rd party Integration with server side auth and shall provide user public documentation, wiki, sandbox and a mock server	D
	FR-112	The system shall provide public RESTful API with JSON with compliance of OPEN API standard for product search, product list and view for live products etc	M
	FR-113	The system shall only expose proxy server (Cloudflare) and entry node IP (NGINX) to public space	M
	FR-114	The system shall block all access to microservices for all available API except Category 1	M
	FR-115	The system shall use InsureTech Internal protocol for IoT based data extraction and data binding	F
	FR-116	The system shall consolidate, annotate, add context and process and save data within regulatory limitation to train internal AI agents	F
	FR-117	The system shall generate statistics, prediction based on big data and provide special projection data to partners given special agreement with InsureTech	F
	FR-118	The system shall implement WebSocket connection for real-time updates	D

4.14 Data Storage

FG	ID	Requirement Description	Priority
FG-014		Data Storage	M
	FR-119	The system shall store all uploaded objects with AWS S3 or equivalent data storage system	M
	FR-120	The system shall store relational data on self-hosted PostgreSQL V17 database (ACID compliance, JSONB support, Bengali full-text search) with Connection pooling: PgBouncer, max 100 connections per service, Read replicas for reporting queries, Query optimization: <100ms for 95% queries, Indexing: Comprehensive indexes on foreign keys, status fields, dates	M
	FR-121	The system shall store finance transaction records, debit-credit records, insurance records and contracts on special purpose built database (like TigerBeetle or equivalent)	D
	FR-122	The system shall use cache database for faster data delivery - Product catalog: 5-minute TTL, User sessions: Redis with 15-minute expiry	M
	FR-123	The system shall process tokenized data on vector database like Pgvector (PostgreSQL) or PineCone	D
	FR-124	The system shall store app native encrypted data in user device in SQLite	M
	FR-125	The system shall store product catalog and metadata, unstructured data in a NoSQL Database (AWS DynamoDB/MongoDB)	M
	FR-126	Upload data policy - Client-side compression: 5MB → 1-2MB (JPEG 80% quality, 1920×1080 max resolution), Chunked upload: 1MB chunks with resume capability (tus.io protocol), Presigned S3 URLs: Direct upload, 30-minute expiry	M
	FR-127	Backup: Daily full, 6-hour incremental, continuous transaction logs	F
	FR-128	Retention: 20 years for regulatory compliance, tiered storage (hot/warm/cold)	F

4.15 Business Rules & Workflows (MD FEEDBACK)

FG	ID	Requirement Description	Priority
FG-015		Business Rules & Workflows	M
	FR-129	Premium Calculation Fallbacks: If insurer API fails, use cached rates (max 24hrs old); if unavailable, queue quote and notify customer within 2 hours	M
	FR-130	Premium Calculation Edge Cases: Handle age-based loading, occupation risk factors, pre-existing conditions with clear messaging	M
	FR-131	Duplicate Policy Detection: Block duplicate policy purchase for same product + same insured person within 30 days; allow cross-product purchases	M
	FR-132	Policy Merge Workflow: Focal Person can merge duplicate accounts after verifying NID, transfer policies, consolidate claims history	M
	FR-133	Claim Status State Machine: Define explicit states (Submitted → Under Review → Documents Requested → Approved/Rejected → Payment Initiated → Settled/Closed)	M
	FR-134	Claim Status Transition Rules: Auto-move to "Documents Requested" if incomplete; require Business Admin+Focal Person approval for >BDT 50K	M
	FR-135	Renewal Timeline: Send first reminder 30 days before expiry, second at 15 days, third at 7 days; allow renewal 45 days before to 30 days after expiry	M
	FR-136	Grace Period Logic: 30-day grace period post-expiry with coverage continued; auto-lapse if unpaid after grace period	M
	FR-137	Lapsed Policy Reinstatement: Allow reinstatement within 90 days of lapse with medical underwriting; require Focal Person approval	D

4.16 Integration Details (MD FEEDBACK)

FG	ID	Requirement Description	Priority
FG-016		Integration & API Contracts	M
	FR-138	API Contract Specification: All Category 3 APIs must provide OpenAPI 3.0 spec with request/response schemas, error codes (400-BadRequest, 401-Unauthorized, 404-NotFound, 409-Conflict, 422-ValidationError, 500-ServerError, 503-ServiceUnavailable), example payloads	M
	FR-139	Insurer API Payloads: Premium Calculation API: Request {productId, age, sumAssured, tenure, occupation} → Response {basePremium, loadings[], discounts[], finalPremium, breakdown[]}; Policy Issuance API: Request {quoteId, customerDetails, paymentProof} → Response {policyNumber, certificateUrl, effectiveDate}	M
	FR-140	Payment Gateway Payloads: Initiate Payment: Request {orderId, amount, currency:BDT, customerPhone, returnUrl, webhookUrl} → Response {paymentUrl, transactionId}; Webhook Callback: {transactionId, status, orderId, amount, signature(HMAC-SHA256)}	M
	FR-141	Retry Logic: Failed API calls retry with exponential backoff: 1s, 2s, 4s, 8s, 16s (max 5 retries); Use circuit breaker pattern (open after 5 consecutive failures, half-open after 30s, close after 3 successes)	M
	FR-142	Idempotency: All payment and policy issuance APIs must accept Idempotency-Key header (UUID); Store idempotency keys for 24 hours; Return cached response for duplicate requests within 24hrs	M
	FR-143	Callback Security: Payment gateway webhooks must include HMAC-SHA256 signature in header; System validates signature using shared secret; Reject unsigned/invalid callbacks; Log all callback attempts	M
	FR-144	EHR Integration Approach - Option A (Preferred): Use LabAid FHIR API with Patient resource matching by NID/phone; Query Encounter resources for IPD admissions; Verify policy coverage in real-time; Pre-authorization workflow with Claim resource	D
	FR-145	EHR Integration Approach - Option B (Fallback): Use LabAid custom REST API with endpoints: GET /patients/{nid}/admissions, POST /preauth/verify, GET /bills/{admission_id}; Secure with mutual TLS + API key	M
	FR-146	EHR Integration Timeout: Set connection timeout 5s, read timeout 15s; If timeout, queue for manual verification; Notify hospital staff via SMS	M

4.17 Fraud Detection & Risk Controls (MD FEEDBACK)

FG	ID	Requirement Description	Priority
FG-017		Fraud Detection & Operations Controls	M
	FR-147	Basic Claim Abuse Checks: Flag claims with (1) Submission within 48hrs of policy purchase, (2) Same claim type >2 times in 12 months, (3) Claim amount = policy limit exactly, (4) Medical provider not in approved network	M
	FR-148	Monitoring Ownership - RACI: Responsible: DevOps Engineer (24x7 monitoring), Accountable: System Admin (incident escalation), Consulted: Business Admin (business impact), Informed: All stakeholders (post-incident report)	M
	FR-149	Escalation RACI for P1 Incidents: Responsible: On-call engineer (immediate response), Accountable: System Admin (decision maker), Consulted: Database Admin + Repository Admin (technical expertise), Informed: Business Admin + Focal Person (business continuity)	M
	FR-150	Fraud Detection Dashboard: Business Admin and Focal Person access real-time dashboard showing flagged transactions, AML alerts, duplicate policy attempts, frequent claim patterns with drill-down capability	M

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5. External Interface Requirements

5.1 User Interfaces

Mobile Application Requirements:

Requirement	Specification	Rationale
Screen Design	Consistent with provided prototype designs	Bangladesh-specific UX considerations
Orientation Support	Portrait and landscape	Enhanced user experience
Screen Size Optimization	4.7" to 6.7"	Coverage of common device range
Theme Support	Dark mode and light mode	User preference and battery saving
Touch Targets	Minimum 44px	Accessibility guidelines compliance
Download Size	Initial: 10MB, Maximum: 100MB	Bangladesh network constraints
Offline Mode	Critical data persistent view	Network reliability issues
Voice Assistance	For Type 3 users	Low digital literacy support

Web Portal Requirements:

Requirement	Specification	Rationale
Design Approach	Desktop-first responsive design	Admin/partner primary use case
Resolution Support	1024x768 to 4K displays	Wide device coverage
Multi-tab Support	Multiple browser tabs	Productivity enhancement
Search & Filter	Comprehensive capabilities	Data management efficiency

5.2 API Architecture & Communication Protocols

API Category Structure:

API Category	Protocol	Use Case	Security Layer	Performance Target
Category 1	Protocol Buffer + gRPC	Gateway ↔ Microservices	System Admin Middle Layer	< 100ms
Category 2	GraphQL + JWT	Gateway ↔ Customer Device	JWT + OAuth v2	< 2 seconds
Category 3	RESTful + JSON (OpenAPI)	3rd Party Integration	Server-side Auth	< 200ms
Public API	RESTful + JSON (OpenAPI)	Product Search/List	Public Access	< 1 second

5.3 Third-Party System Integrations

Payment Gateway Integrations:

Provider	Integration Method	Scope	Compliance Requirements
bKash	REST API + Webhook	Payment processing, refunds, transaction status	bKash merchant agreement
Nagad	REST API + Webhook	Payment processing, refunds, transaction status	Nagad partner certification
Rocket	REST API + Webhook	Payment processing, refunds, transaction status	DBBL partnership
Card Processors	PCI-DSS Gateway	Credit/debit card processing	PCI-DSS Level 1 compliance

Insurance Company API Integration:

- Premium calculation and underwriting APIs
- Policy issuance and certificate generation
- Claims processing and settlement APIs
- Regulatory reporting data exchange
- Real-time policy status updates

LabAid EHR System Integration:

- Cashless claim verification for IPD services
- Pre-authorization workflows
- Medical record access (with consent)
- Integration via secured HL7/FHIR endpoints or custom APIs
- Timeout handling: 5s connection, 15s read

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6. Non-Functional Requirements

ID	Requirement Description	Priority
NFR-001	The system shall ensure 99.5% uptime to meet critical business needs	M
NFR-002	The system shall encrypt (TLS 1.3 and AES-256) all sensitive data in transit and at rest	M
NFR-003	The system shall handle up to 1,000 concurrent users without performance degradation	M
NFR-004	The system shall handle up to 5,000 concurrent users without performance degradation	D
NFR-005	The system shall handle up to 10,000 concurrent users without performance degradation	F
NFR-006	The system shall have a median response time of less than 1 second and not more than 2s for 95% of user actions	M
NFR-007	Disaster Recovery: RTO < 15 minutes for critical services, RPO < 1 hour	M
NFR-008	The system shall have Category 1 API response time < 100ms	D
NFR-009	The system shall have Category 3 API response time < 200ms	D
NFR-010	The system shall have Category 2 API response time < 2 sec	D
NFR-011	The system shall have public API response time < 1 sec	D
NFR-012	The system shall ensure minimum app startup time < 5 sec	D
NFR-013	The system shall support 1 million active users and 200K active policies in 24 months with auto vertical scaling	F
NFR-014	The system shall have automatic switch option for user to different mobile device with data migration	F
NFR-015	The system shall have multi-language support (English + Bengali)	D

7. Data Model & Storage Requirements

7.1 Database Architecture Distribution

Storage System	Use Case	Performance Target	Rationale
PostgreSQL V17	Relational data, user/policy/claims	< 100ms for 95% queries	ACID compliance, JSONB, Bengali full-text search
TigerBeetle	Financial transactions, insurance records	< 10ms transaction time	Purpose-built for financial accuracy
Redis	Cache, sessions	< 1ms access time	Product catalog (5min TTL), User sessions (15min)
DynamoDB/MongoDB	Product catalog, metadata	< 50ms read time	NoSQL flexibility for unstructured data
AWS S3	Documents, uploaded objects	< 5s upload/download	Scalable object storage with encryption
Pgvector/PineCone	Tokenized data, AI/ML	< 100ms similarity search	Vector database for ML operations
SQLite	Mobile app local data	< 10ms local query	Encrypted offline capability

7.2 Enhanced Data Entities (MD FEEDBACK)

Nominee/Beneficiary Entity:

```
NomineeID (Primary Key, UUID)
PolicyID (Foreign Key)
FullName (VARCHAR 200, NOT NULL)
Relationship (ENUM: Spouse, Child, Parent, Sibling, Other)
DateOfBirth (DATE)
NationalID (VARCHAR 20, UNIQUE)
SharePercentage (DECIMAL 5,2, CHECK >0 AND <=100)
ContactPhone (VARCHAR 15)
Address (TEXT)
CreatedAt, UpdatedAt
CONSTRAINT: SUM(SharePercentage) per PolicyID = 100
```

KYCDocument Entity:

```
DocumentID (Primary Key, UUID)
UserID (Foreign Key)
DocumentType (ENUM: NID_Front, NID_Back, Passport, Photo, Utility_Bill,
Income_Proof, Medical_Report)
FileName (VARCHAR 255)
```

```
S3Path (TEXT)
FileHash (SHA256, for integrity)
UploadedAt (TIMESTAMP)
VerificationStatus (ENUM: Pending, Verified, Rejected)
VerifiedBy (Foreign Key → Stakeholder)
VerifiedAt (TIMESTAMP)
RejectionReason (TEXT, NULL)
OCRData (JSONB, extracted fields)
ExpiryDate (DATE, for passports/utility bills)
```

Quote Entity:

```
QuoteID (Primary Key, UUID)
UserID (Foreign Key, NULL if anonymous)
ProductID (Foreign Key)
RequestedSumAssured (DECIMAL)
RequestedTenure (INT, years)
CustomerAge (INT)
Occupation (VARCHAR 100)
BasePremium (DECIMAL)
Loadings (JSONB, {reason: amount}[])
Discounts (JSONB, {code: amount}[])
FinalPremium (DECIMAL)
QuoteValidUntil (TIMESTAMP, +48hrs)
ConvertedToPolicyID (Foreign Key, NULL)
CreatedAt, UpdatedAt
```

PartnerCommission Entity:

```
CommissionID (Primary Key, UUID)
PartnerID (Foreign Key)
PolicyID (Foreign Key)
CommissionType (ENUM: Acquisition, Renewal, Claims_Assistance)
CommissionRate (DECIMAL 5,2, percentage)
CommissionAmount (DECIMAL)
PayoutStatus (ENUM: Pending, Approved, Paid, Withheld)
PayoutDate (DATE, NULL)
PayoutReference (VARCHAR 100, payment txn ID)
CreatedAt, UpdatedAt
```

NotificationLog Entity:

```
NotificationID (Primary Key, UUID)
UserID (Foreign Key, NULL for system-wide)
Channel (ENUM: SMS, Email, Push, In_App)
EventType (ENUM: OTP, Purchase_Confirmation, Claim_Update, Renewal_Reminder,
Marketing)
MessageTemplate (VARCHAR 50, template ID)
MessageContent (TEXT)
RecipientPhone (VARCHAR 15, for SMS)
RecipientEmail (VARCHAR 255, for Email)
DeviceToken (TEXT, for Push)
SendStatus (ENUM: Queued, Sent, Failed, Delivered, Bounced)
SentAt (TIMESTAMP)
DeliveredAt (TIMESTAMP, NULL)
FailureReason (TEXT, NULL)
RetryCount (INT, DEFAULT 0)
CreatedAt
```

7.3 Data Storage Distribution Strategy

Data Type	Storage System	Rationale	Performance Target
User & Policy Data	PostgreSQL V17	ACID compliance, complex queries, Bengali support	< 100ms query time
Financial Transactions	TigerBeetle	Purpose-built for financial accuracy, double-entry bookkeeping	< 10ms transaction time
Product Catalog	DynamoDB/MongoDB	Flexible schema, fast reads	< 50ms read time
Documents & Files	AWS S3	Scalable object storage, encryption at rest	< 5s upload/download
Cache & Sessions	Redis	Fast in-memory operations	< 1ms access time
Vector/AI Data	Pgvector/PineCone	ML model data, similarity search	< 100ms similarity search
Mobile App Data	SQLite	Offline capability, encrypted local storage	< 10ms local query

8. Security & Compliance Requirements

8.1 Security Infrastructure & Key Management

ID	Requirement Description	Priority
SEC-001	The system shall use separate secret vault - AWS KMS/Azure Key Vault/HashiCorp, 90-day key rotation	M
SEC-002	The system shall use Data Masking: NID (last 3 digits), phone (mask middle), email (mask username)	M
SEC-003	The system shall follow PCI-DSS compliance for card flows - Approach: Hosted payment page (redirect model) - DO NOT store card data, Level: SAQ-A (simplest, for redirecting merchants), Requirements: Annual SAQ, quarterly ASV scans, TLS 1.3, Tokenization: Store only gateway tokens for recurring payments	M
SEC-004	The system shall have AML/CFT detection hooks - Transaction Monitoring: 20+ automated rules for AML detection including Rapid purchases (>3 policies in 7 days), High-value premiums (>BDT 5 lakh), Frequent cancellations, Mismatched nominees, Geographic/payment anomalies	D
SEC-005	The system shall have IDRA reporting capabilities following IDRA data format - Monthly Reports: Premium Collection (Form IC-1), Claims Intimation (Form IC-2), Quarterly Reports: Claims Settlement (IC-3), Financial Performance (IC-4), Annual Reports: FCR (Financial Condition Report), CARAMELS Framework Returns, Event-Based: Significant incidents (48hrs), fraud cases (7 days), Platform: Report generator with IDRA Excel templates, audit trail, 20-year archive	D
SEC-006	The system shall have regular penetration testing - Penetration Testing: Pre-launch + annually (SISA InfoSec or international firm)	D
SEC-007	The system shall have regular security audits from various security auditors and regulatory bodies and maintain compliance	D
SEC-008	DAST: OWASP ZAP/Burp Suite (weekly on staging)	D
SEC-009	SAST: SonarQube/Checkmarx (every commit, block critical vulnerabilities)	D
SEC-010	Virus scanning: ClamAV on uploaded files	M

8.2 Enhanced IDRA Compliance (MD FEEDBACK)

ID	Requirement Description	Priority
SEC-011	IDRA Monthly Reports: Generate Form IC-1 (Premium Collection) by 10th of each month with breakdown by product line, geographic region, partner channel in Excel format per IDRA template v2024	M
SEC-012	IDRA Monthly Reports: Generate Form IC-2 (Claims Intimation) by 10th of each month listing all new claims with policy number, claim amount, claim type, date of intimation	M
SEC-013	IDRA Quarterly Reports: Generate Form IC-3 (Claims Settlement) within 15 days of quarter-end showing settlement ratio, average TAT, pending >30 days breakdown	M
SEC-014	IDRA Quarterly Reports: Generate Form IC-4 (Financial Performance) within 20 days of quarter-end with premium earned, claims paid, commission paid, net profit/loss	M
SEC-015	IDRA Annual FCR: Generate Financial Condition Report (FCR) within 90 days of year-end including full CARAMELS framework assessment with external auditor sign-off	M
SEC-016	IDRA Event-Based Reporting: Report significant incidents (fraud >BDT 1L, data breach, system outage >4hrs) within 48 hours via IDRA portal	M

8.3 Enhanced AML/CFT Compliance (MD FEEDBACK)

ID	Requirement Description	Priority
SEC-017	AML/CFT Concrete Triggers: Flag transactions matching: (1) >3 policies in 7 days, (2) Premium >BDT 5L without income proof, (3) Nominee mismatch with no relationship doc, (4) Payment from third-party account, (5) Frequent cancellations >2 in 30 days, (6) Geographic anomaly (policy in Dhaka, payment from remote district), (7) Multiple failed KYC attempts >3, (8) PEP match in screening	M
SEC-018	SAR Workflow: (1) System auto-flags suspicious transaction → (2) Compliance Officer reviews within 24hrs → (3) If confirmed suspicious, escalate to Business Admin+Focal Person → (4) Prepare SAR with evidence → (5) Submit to BFIU within 3 business days → (6) Mark account for enhanced monitoring → (7) Do NOT notify customer (tipping off prohibited)	M
SEC-019	Data Deletion Exceptions: Customer data deletion requests processed within 30 days EXCEPT: (a) Active policy holders (deletion after policy expiry+7yrs), (b) Ongoing claims (deletion after settlement+7yrs), (c) Under SAR investigation (deletion prohibited until case closed), (d) Regulatory hold (deletion requires IDRA/BFIU approval)	M
SEC-020	Right to Erasure Workflow: Customer submits deletion request → System validates exceptions → If eligible, anonymize PII while retaining transaction records → Generate deletion certificate → Notify customer within 30 days	D

8.4 Data Protection & Encryption Standards

Data Classification	Encryption Standard	Key Management	Access Control
Personally Identifiable Information (PII)	AES-256	AWS KMS with 90-day rotation	Role-based with audit logging
Financial Transaction Data	AES-256 + Additional Hashing	TigerBeetle built-in encryption	Restricted access with MFA
KYC Documents	AES-256 with client-side encryption	End-to-end encryption	Compliance officer access only
Medical Records	AES-256 with additional anonymization	Healthcare-specific key management	Medical staff + consent-based
Audit Logs	AES-256 with immutable storage	Centralized key management	Read-only access for auditors

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9. Performance & Scalability Requirements

9.1 Performance Benchmarks

Metric	Baseline Target	Peak Load Target	Measurement Method
Category 1 API (gRPC)	< 100ms	< 150ms	APM tools (New Relic/Datadog)
Category 2 API (GraphQL)	< 2 seconds	< 3 seconds	GraphQL monitoring
Category 3 API (REST)	< 200ms	< 300ms	API gateway monitoring
Public API	< 1 second	< 1.5 seconds	Public endpoint monitoring
Mobile App Startup	< 5 seconds	< 7 seconds	Device testing
PostgreSQL Query	< 100ms for 95%	< 150ms for 95%	Database monitoring
TigerBeetle Transaction	< 10ms	< 20ms	Financial system monitoring

9.2 Capacity Planning

Component	Current Capacity	12-Month Target	24-Month Target	Scaling Strategy
Concurrent Users	1,000	5,000	10,000	Auto-scaling with CloudWatch metrics
API Requests/Second	100	1,000	5,000	gRPC microservices scaling
Database Connections	100 (PostgreSQL)	500	2,000	PgBouncer connection pooling
TigerBeetle TPS	1,000	10,000	50,000	TigerBeetle cluster scaling
Storage (TB)	1	10	50	Auto-scaling object storage
Policy Documents	10,000	500,000	2,000,000	Distributed storage with archival

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10. AML/CFT Compliance Requirements

10.1 Customer Due Diligence (CDD) Framework

Mandatory CDD Requirements for Bangladesh:

Requirement	Implementation	Compliance Standard
Identity Verification	NID/Passport verification via approved eKYC	BFIU Guidelines
Address Verification	Utility bill or bank statement	MLPA Requirements
Photo Identification	Selfie with liveness detection	Enhanced CDD
Source of Funds	Income declaration for high-value policies	Risk-based approach
PEP Screening	Automated screening against watchlists	FATF Recommendations

10.2 Risk-Based Customer Categorization

Risk Level	Criteria	CDD Requirements	Monitoring Frequency
Low Risk	Standard customers, low premium policies	Standard CDD	Annual review
Medium Risk	Higher premiums, multiple policies	Enhanced documentation	Quarterly review
High Risk	PEPs, large premiums, suspicious patterns	Enhanced Due Diligence (EDD)	Monthly monitoring
Prohibited	Sanctioned individuals, blocked entities	Transaction rejection	Real-time blocking

10.3 Automated AML Monitoring Rules

Monitoring Rule	Threshold	Alert Level	Action Required
Rapid Policy Purchases	>3 policies in 7 days	High	Enhanced verification
High-Value Premiums	>BDT 5 lakh	High	Management approval
Frequent Cancellations	>2 cancellations in 30 days	Medium	Pattern analysis
Mismatched Nominees	Different family names without relationship proof	Medium	Additional documentation
Geographic Anomalies	Transaction from unusual location	Low	Location verification
Payment Method Inconsistency	Different mobile numbers vs NID	Medium	Customer verification

10.4 Record Keeping & Audit Trail

Document Type	Retention Period	Storage Requirements	Access Controls
CDD Documentation	5+ years after relationship end	Encrypted PostgreSQL + S3	Compliance team only
Transaction Records	7+ years	TigerBeetle + Archive	Audit and compliance
STR Documentation	10+ years	Secured offline storage	Business Admin + Focal Person
Training Records	5+ years	HR system integration	HR and compliance
System Audit Logs	20+ years	Immutable PostgreSQL logging	System administrators

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11. Operational Requirements & Support

11.1 System Monitoring & Alerting

Monitoring Category	Metrics	Alert Thresholds	Response Time
Application Health	gRPC/GraphQL response times, error rates	>100ms (gRPC), >2s (GraphQL), >1% error rate	5 minutes
Infrastructure	CPU, memory, disk usage	>80% utilization	10 minutes
Database Performance	PostgreSQL query time, TigerBeetle TPS	>100ms queries, <1000 TPS	5 minutes
Security Events	Failed logins, privilege escalation	>10 failed attempts	Immediate
Business Metrics	Policy sales, claim processing	<50% of daily target	1 hour

11.2 Incident Management Framework

Priority Level	Definition	Response Time	Escalation
P1 - Critical	System down, data loss, security breach	15 minutes	Immediate management notification
P2 - High	Major feature unavailable	1 hour	Team lead notification
P3 - Medium	Minor feature issues	4 hours	Standard queue processing
P4 - Low	Cosmetic issues, enhancement requests	24 hours	Next business day

11.3 Support Structure

Support Level	Scope	Availability	Response SLA
Tier 1 - Self-Service	FAQ, knowledge base, AI chatbot	24x7	Immediate
Tier 2 - Call Center	General inquiries, account issues	Business hours	2 minutes
Tier 3 - Technical Support	Complex issues, escalations	Business hours	1 hour
Tier 4 - Engineering	System bugs, critical issues	On-call rotation	30 minutes

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12. Acceptance Criteria & Test Summary

12.1 Critical Business Workflow Validation

Workflow	Acceptance Criteria	Success Metrics
User Registration	Phone-based registration with OTP validation completes successfully	>95% completion rate
KYC Verification	Document upload and verification process completes within 5 minutes	>90% automated approval
Policy Purchase	End-to-end purchase flow from product selection to policy issuance	>99% transaction success
Payment Processing	Multiple payment methods with real-time confirmation	>99.5% payment success
Claim Submission	Claim initiation with document upload and status tracking	<3 minutes submission time
Policy Renewal	Automated and manual renewal workflows	>95% renewal completion

12.2 API Performance Testing

API Category	Load Profile	Success Criteria	Expected Performance
Category 1 (gRPC)	1000 concurrent requests	<100ms response time	High-throughput internal communication
Category 2 (GraphQL)	500 concurrent requests	<2s response time	Mobile-optimized data fetching
Category 3 (REST)	100 concurrent requests	<200ms response time	Standard 3rd party integration
Public API	50 concurrent requests	<1s response time	Public product search

12.3 FR → Test Case Mapping (MD FEEDBACK)

FR-ID	Test Case ID	Test Scenario	Expected Result	Test Type
FR-001	TC-001	Valid Bangladesh phone registration	OTP sent within 60s	Integration
FR-004	TC-002	Duplicate NID registration attempt	Error message displayed	Functional
FR-033	TC-003	End-to-end purchase with bKash	Policy issued within 30s	E2E
FR-051	TC-004	Joint approval (BizAdmin+Focal)	Claim approved only after both	Workflow
FR-129	TC-005	Insurer API failure during quote	Cached rate used + customer notified	Resilience

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13. Traceability Matrix & Change Control

13.1 Requirements Traceability

Business Objective	Related Functional Requirements	Success Metrics
Digital Onboarding: 40,000 policies by 2026	FR-001 to FR-016, FR-033 to FR-040	Monthly policy acquisition rate
API Performance Optimization	FR-107 to FR-118, NFR-008 to NFR-011	API response time metrics
Financial Transaction Integrity	FR-121 (TigerBeetle), SEC-003 (PCI-DSS)	Transaction accuracy and speed
Regulatory Compliance	SEC-011 to SEC-020 (IDRA/AML/CFT)	Audit compliance score
Partner Management Excellence	FR-011 (Focal Person), FR-086 to FR-092	Number of active partners
Claims Efficiency	FR-041 to FR-058, FR-133 to FR-137	Average claim TAT

13.2 Change Control Process

Approval Hierarchy:

15. 1. Dev submits change request
16. 2. Repository Admin reviews code changes
17. 3. Database Admin reviews data model impact
18. 4. System Admin reviews infrastructure impact
19. 5. Business Admin approves business impact
20. 6. Focal Person approves partner-related changes

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14. Appendices

14.1 API Architecture Diagram

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│                        CLOUDFLARE PROXY                      │
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│                        NGINX GATEWAY                         │
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┌──────────────────────────┬───────────────────────────────────┐
│  PUBLIC API (REST/JSON)  │ CATEGORY 2 (GraphQL + JWT)        │
│  - Product Search        │ - Customer Device                 │
│  - Product List          │ - Mobile Apps                     │
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                          │
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                  │  (OAuth2 + JWT)  │
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│CAT 3    │  │      CATEGORY 1      │  │   INTERNAL       │
│REST API │  │  (gRPC + ProtoBuf)   │  │  MICROSERVICES   │
│3rd      │  │  Microservices       │  │   (Kafka         │
│Party    │  │  Communication       │  │   Orchestrated)  │
└─────────┘  └──────────────────────┘  └──────────────────┘
```

14.2 Stakeholder Hierarchy

```
System Admin (Cloud/IAM Root)
    │
    ├── Repository Admin (Code/Deploy)
    │
    ├── Database Admin (Data Management)
    │
    ├── Business Admin (Business Operations)
    │   │
    │   └── Focal Person (Partner Bridge) ★ KEY ROLE
    │       │
    │       └── Partner Admin (Tenant Root)
    │           │
    │           ├── Insurer Agents (Partner Staff)
    │           └── Insurer Underwriters (API Users)
    │
    ├── Dev (Development)
    │
    └── Support/Call Centre (Customer Assistance)
```

14.3 Claims Approval Workflow

```
Customer Submits Claim
    │
    ▼
Partner Agent Initial Review
    │
    ▼
Amount-Based Routing
    │
    ├─ BDT 0-10K ──► L1 Auto/Officer ──► Auto-Approve (24hrs)
    │
    ├─ BDT 10K-50K ──► L2 Manager ──► Approval (3 days)
    │
    ├─ BDT 50K-2L ──► L3 Head ──► Joint Approval (7 days)
    │                                │
    │                                ├── Business Admin ─┐
    │                                │                   ├─► Decision
    │                                └── Focal Person ───┘
    │
    └─ BDT 2L+ ──► Board + Insurer ──► Final Approval (15 days)
                          │
                          ▼
                    Payment Processing
```

14.4 Technology Stack

Component	Technology	Rationale
Financial DB	TigerBeetle	Purpose-built for financial accuracy
Primary DB	PostgreSQL V17	ACID compliance, Bengali support
Category 1 API	gRPC + Protocol Buffers	High-performance microservices
Category 2 API	GraphQL	Efficient mobile data fetching
Category 3 API	REST + OpenAPI	3rd party integration standard
Event Orchestration	Kafka	Event-driven architecture
Cache	Redis Cluster	Fast in-memory operations
Object Storage	AWS S3	Scalable document storage
Monitoring	Datadog/Prometheus	Comprehensive observability
CDN/Proxy	Cloudflare	DDoS protection

Document Approval & Sign-off

Technical Architecture Confirmation:

By signing below, stakeholders confirm their acceptance of this System Requirements Specification V2.2 including:

- **Function Groups (FG-001 to FG-017)** with incremental Functional Requirements (FR-001 to FR-150)
- **SEC numbering (SEC-001 to SEC-020)** for security requirements
- **API Architecture:** Category 1 (gRPC), Category 2 (GraphQL), Category 3 (REST), Public (REST)
- **Database Strategy:** PostgreSQL V17 + TigerBeetle + Redis + DynamoDB/MongoDB + S3 + SQLite
- **Novel Features:** Focal Person role, Joint Approval, WebRTC video calls, Kafka orchestration
- **User Categorization:** Type 1 (Urban), Type 2 (Semi-Urban), Type 3 (Rural/Voice-assisted)
- **Mobile Constraints:** Offline mode, Low bandwidth mode, Gradual download (10MB → 100MB)
- **MD Feedback Integrated:** Business rules, IDRA reports, API contracts, Enhanced data model

Role	Name	Signature	Date
System Admin	_____	_____	_____
Repository Admin	_____	_____	_____
Database Admin	_____	_____	_____
Business Admin	_____	_____	_____
Focal Person	_____	_____	_____
Compliance Officer	_____	_____	_____

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